

Adversarial Vulnerability of Machine Learning GNSS Spoofing Detectors to Transmitted Spoofing Signals

Abstract

A spoofer broadcasts counterfeit Global Navigation Satellite System (GNSS) signals so that a victim receiver computes a false position or time. Detectors trained by machine learning are increasingly proposed for this threat, but are evaluated against fixed corpora of recorded attacks that do not respond to the detector under test. This paper evaluates fourteen detectors, eight classical and six deep, under attacks that are radio signals selected by an adversary. A counterfeit GPS L1 C/A waveform is synthesised at sample level, summed with authentic recordings from the Texas Spoofing Test Battery and tracked by the FGI-GSRx software receiver, so that the observables each detector reads are produced by receiver processing of a transmittable signal. The adversary sets counterfeit power, code pull-off rate and carrier frequency and phase, with no knowledge of the detector and no query access. The generator is validated by reproducing data set ds7, whose construction is published. All detectors are placed at a common operating point flagging 95 percent of spoofed samples on clean validation data, and evaluated against 126 settings satisfying the published conditions for receiver takeover. Clean performance predicts resistance poorly and the two orderings are close to inverted: the random forest attains the highest clean F_1 score of 0.833 and the lowest false alarm rate of 0.337, yet is evaded by 55 of the 126 attacks. Detectors evading none flag 46 to 74 percent of authentic data. One evaded attack drives the receiver clock at 48.89 nanoseconds per second against a commanded 48.87.

Keywords: GNSS spoofing, Spoofing detection, Machine learning, Software defined receiver, Resilient PNT, Receiver in the loop testing

1 Introduction

1.1 GNSS spoofing and its detection

A Global Navigation Satellite System (GNSS) supplies a receiver with its position, its velocity and its time. Power grids use that time to synchronise, financial systems use it to stamp transactions, and aircraft and autonomous platforms use the position to navigate ([Kaplan and Hegarty 2005](#)). The civil signal that carries this service is public and unencrypted, and its structure is fully specified in open documents, so a

transmitter of modest cost can generate a waveform with the same modulation, the same spreading code and the same message format as a genuine satellite. A spoofer is such a transmitter. It emits counterfeit satellite signals with the intention that a victim receiver tracks them in preference to the authentic ones and reports a position or a time of the spoofer’s choosing. The threat is not hypothetical: a spoofer has steered a yacht off its intended course through the ship’s navigation system (Bhatti and Humphreys 2017), and a spoofer has captured an airborne platform in flight (Kerns et al. 2014).

Spoofing differs from jamming in a way that determines how it must be handled. A jammer denies the service, and the receiver observes that it has lost lock. A spoofer leaves the receiver operating nominally while the solution it reports is wrong, so the receiver holds no internal evidence that anything has occurred and the attack must be identified while it is in progress (Psiaki and Humphreys 2016). Capturing a receiver that is already tracking authentic signals is correspondingly more demanding than simply transmitting at higher power. The published procedure proceeds in three stages: the counterfeit is first aligned with the authentic signal in code phase, carrier phase and Doppler frequency, so that the victim’s correlator sees a single undistorted peak; its power is then raised slightly above the authentic power, typically by a few decibels; and only then is the counterfeit dragged away from the authentic signal, carrying the tracking loops with it and moving the reported position or time (Psiaki and Humphreys 2016; Kerns et al. 2014). A counterfeit that omits the alignment stage does not capture the receiver at all, and acts only as interference. The consequence is that the attacker’s freedom is bounded: not every combination of transmitter settings takes over a receiver, and an evaluation that ignores this distinction credits a detector for rejecting transmissions that would never have succeeded.

Classical anti-spoofing tests monitor quantities the receiver computes in the course of tracking a satellite. Representative examples are the carrier-to-noise density ratio, which reflects received signal strength; the shape of the correlation function, which distorts when a second copy of the same code is present; and the agreement between code and carrier measurements of a single satellite (Akos 2012; Psiaki and Humphreys 2016). Each such test identifies one physical signature of an attack, which makes it interpretable and also makes it evadable: a spoofer that holds the named quantity within its nominal range passes the test, and the set of tests a receiver runs is public knowledge.

1.2 Learned detectors and how they are evaluated

The limitation of signature based tests has motivated detectors inferred from data. A machine learning detector is a classifier trained on labelled epochs, some recorded while the receiver tracked authentic signals and some recorded under spoofing, which learns which joint configurations of receiver measurements indicate an attack. Because such a classifier reads many measurements simultaneously, it can exploit dependencies that no single hand written statistic encodes, and recent work reports high detection accuracy for both deep architectures and classical classifiers applied to receiver observables (Borhani-Darian et al. 2024; Li et al. 2025; Chen et al. 2025). Physics constrained networks extend the approach towards attacks absent from the training data by imposing relations that authentic signals must satisfy (Song et al. 2026).

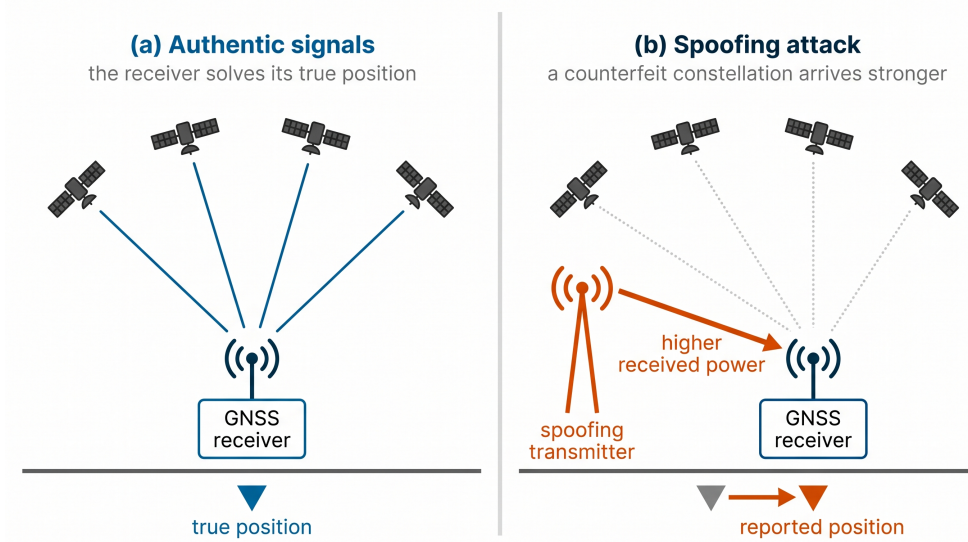


Fig. 1 A GNSS receiver under authentic signals and under a spoofing attack. In (a) the receiver solves its true position from the satellite signals. In (b) a spoofing transmitter emits a counterfeit constellation that arrives at higher received power, captures the tracking loops, and moves the reported position while the receiver continues to report a valid solution.

The evaluation practice that accompanies this literature is uniform: a detector is scored against a fixed collection of recorded or simulated attacks. Those attacks were selected before the detector existed, by whoever assembled the corpus, and they do not respond to the classifier under test. A spoofer does respond. It selects the counterfeit power, the rate at which it drags the code away from the authentic code, and the frequency and phase of its carrier, and each of those choices alters the measurements the victim receiver produces and therefore alters what the detector reads. Detection accuracy obtained against a fixed corpus describes the corpus. It does not, by itself, characterise behaviour against an adversary that chooses what to transmit, and the two quantities need not coincide.

1.3 Receiver in the loop evaluation of transmitted attacks

This paper evaluates fourteen detectors under attacks that are radio signals rather than perturbations of a measurement table. A counterfeit GPS L1 C/A waveform is synthesised at sample level, summed with authentic recorded samples from the Texas Spoofing Test Battery (Humphreys et al. 2012; Humphreys 2016), passed through a model of the receiver front end, and tracked by the FGI-GSRx software receiver (Finnish Geospatial Research Institute 2022). The observables each detector reads are those the receiver computes while tracking that sum. Figure 1 contrasts authentic operation with a spoofing attack, and Figure 2 shows the evaluation chain.

Evaluating at the waveform level fixes the relationship between the adversary’s actions and the detector’s inputs. The adversary acts on transmitter settings, listed

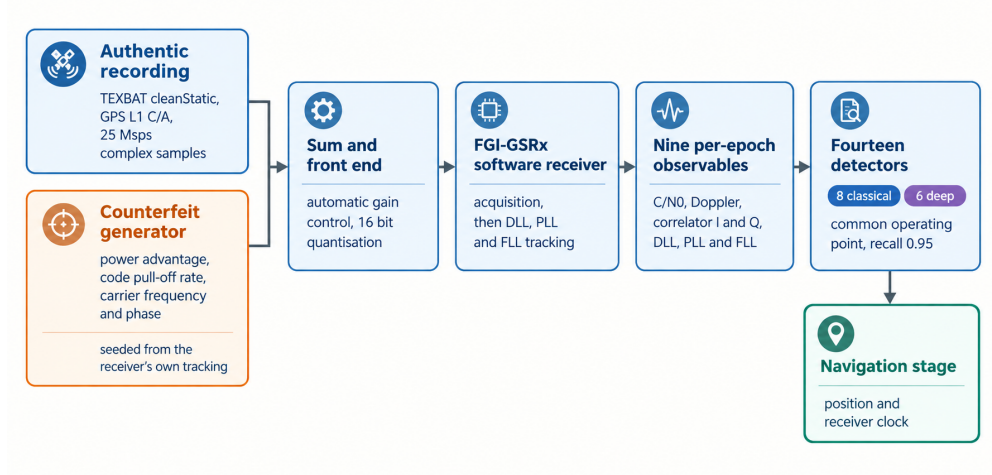


Fig. 2 How a transmitted attack becomes a detector input. A counterfeit GPS L1 C/A signal is generated as radio samples and added to authentic recorded samples. The sum passes through the front end and is tracked by the FGI-GSRx software receiver, which produces the nine per-epoch measurements the detectors read. The same tracking output also feeds the navigation stage, which yields the position and the receiver clock.

in Table 2, which are the quantities a physical spoofer controls; the observables follow from a receiver tracking the resulting signal, and inherit whatever dependencies receiver processing imposes on them. Several of the observables a detector reads are computed from the same underlying correlator output and therefore cannot be varied independently, so the mapping from adversary action to detector input is not the identity and is not free. The generator is validated against data set ds7 of the battery, a recorded attack whose construction is documented in full (Humphreys 2016), and reproduces the observables recorded for it.

All fourteen detectors are placed at a common operating point, defined as the highest decision threshold at which a detector still flags 95 percent of spoofed samples on clean validation data. A common operating point is necessary because a classifier’s error rates move together with its threshold, so detectors read at different thresholds are not comparable. The false alarm rate each detector pays to meet the recall floor is reported alongside every result.

Three findings follow. First, ranking on clean data inverts under transmitted attack: the random forest attains the highest clean F_1 score, the harmonic mean of precision and recall, and the lowest false alarm rate, yet fails to flag 55 of the 126 transmitted attacks that capture the receiver, flagging as few as 0.1 percent of the spoofed samples of one of them. Second, the detectors that flag every attack achieve this only by flagging 46 to 74 percent of authentic data, a regime in which a receiver would alarm almost continuously, so their apparent resistance is not operationally available. Across all fourteen, false alarm rates at this operating point span 34 to 96 percent. Third, a missed attack propagates to the navigation solution: followed through the navigation stage, one such attack drives the receiver clock at 48.89 nanoseconds per second against

a commanded 48.87, after the receiver’s own drift of 0.12 nanoseconds per second has been measured and removed.

1.4 Contributions

- An evaluation methodology for GNSS spoofing detectors in which every attack is synthesised as radio samples, summed with authentic recordings and tracked by a software receiver, so that the detector inputs are produced by receiver processing of a transmittable waveform. The generator is validated by reproducing a recorded attack whose construction is published.
- A library of 209 transmitter settings spanning counterfeit power, code pull-off rate and carrier alignment, of which 126 satisfy the published conditions for receiver takeover. The library expresses the adversary’s choice set as data and separates transmissions that capture a receiver from those that do not.
- A comparative evaluation of fourteen detectors, eight classical and six deep, against that library at a common operating point, reporting for each detector the number of transmitted attacks it fails to flag together with the false alarm rate it incurs on authentic data.
- A navigation domain measurement establishing that a missed attack reaches the position and timing solution, obtained by propagating one such attack through the navigation stage and comparing the induced clock drift against the rate the transmitter was configured to impose.

1.5 Organisation

Section 2 reviews spoofing detection and the robustness of learned classifiers. Section 3 describes the receiver processing chain and the observables it produces. Section 4 defines the threat model and the attack space. Section 5 presents the waveform generator, its validation, the attack library, the detector set and the operating point calibration. Section 6 reports the results, Section 7 examines their implications for receiver integrity and states the scope of the study, and Section 8 concludes.

2 Related Work

2.1 Classical spoofing detection

Classical anti-spoofing tests operate on quantities the tracking loops already produce, which makes them inexpensive to compute and straightforward to interpret. Received power monitoring compares the carrier-to-noise density ratio C/N_0 against the value expected for a satellite at a given elevation; a counterfeit transmitted above the authentic power raises this quantity, and the additional energy also registers in the automatic gain control setting of the front end (Akos 2012). Correlation shape tests examine the peak formed when the incoming signal is correlated against a locally generated replica of the spreading code. An authentic signal yields a symmetric peak whose width is set by the chip duration and the front end bandwidth, whereas a counterfeit present alongside the authentic signal superimposes a second peak and distorts the composite, producing asymmetry that can be quantified from early, prompt and late correlator

taps (Psiaki and Humphreys 2016). Consistency tests exploit relations that must hold among measurements of a genuine signal: the code and carrier observables of one satellite track the same changing geometric range and must therefore evolve together, so a counterfeit that displaces one without the other violates the relation (Psiaki and Humphreys 2016).

These tests are complementary in the sense that no one of them subsumes another, and a receiver running several raises the cost of a successful attack. They nonetheless share a structural property. Each is defined by a single named physical signature, which is what makes it interpretable, and a spoofer that maintains the named quantity within its nominal range satisfies the test. Because the family of deployed tests is documented in the open literature, an adversary can treat the signature set as known and design a transmission that respects it. This observation motivates detection rules that are not specified in advance by a human.

2.2 Machine learning based spoofing detection

A learned detector replaces a hand specified statistic with a decision rule inferred from labelled data. A receiver is run over recordings in which the ground truth is known, each epoch yields a feature vector of the observables available at that instant together with a label, and a classifier is fitted to reproduce those labels before being applied to unseen epochs. Because the classifier consumes the observables jointly, it can respond to configurations that no univariate statistic encodes.

Two model families recur in this literature, and both are represented in the present study. Single epoch classifiers operate on one observation vector at a time. Decision trees partition the feature space by recursive threshold tests, and ensembles such as random forests and gradient boosted trees aggregate many such partitions to reduce variance. Nearest neighbour classifiers assign a label from the closest training examples, and support vector machines construct a maximum margin separating surface, in the nonlinear case through a kernel. Sequence models instead consume a window of consecutive epochs and can therefore exploit temporal structure. Convolutional networks extract local patterns across the window, recurrent architectures such as the long short term memory propagate state along it (Hochreiter and Schmidhuber 1997), and attention based architectures weight the epochs of the window against one another (Vaswani et al. 2017).

Applications to GNSS spoofing span both families. Deep networks have been applied to correlator outputs (Borhani-Darian et al. 2024), to signal quality features on small uncrewed aircraft (Sung et al. 2022), to receiver exchange format records (Romaniuc et al. 2025) and to unmanned aerial platforms (Yang et al. 2025). Classical models have been applied through support vector machines (Chen et al. 2022), nearest neighbour classifiers (Mahroof et al. 2024) and feed forward networks benchmarked against recurrent alternatives (Jullian et al. 2022). Real time operation has been addressed (Li et al. 2025), as has enlargement of the training distribution by generative augmentation (Chen et al. 2025). Reported detection accuracy across this body of work is consistently high, and the architectures achieving it are heterogeneous, which suggests that the reported figures are substantially determined by the evaluation corpora rather than by architectural choices.

2.3 Generalization beyond the training attacks

A classifier fitted to recorded examples characterises the attacks contained in those recordings, and whether its accuracy persists on an attack outside them is a separate question that has received direct attention.

One response treats detection as one class learning, fitting only the authentic distribution so that any sufficiently atypical observation is flagged, which removes the dependence on a catalogue of attack examples (Iqbal et al. 2024). A second augments the training set with synthetically generated attacks in order to widen the distribution the detector has seen (Chen et al. 2025). A third constrains the detector with physics: code carrier consistency, the same relation the classical consistency test evaluates, has been imposed as a training constraint so that the network is required to respect a relation authentic signals satisfy, and this is reported to preserve accuracy across held out scenarios on which unconstrained detectors degrade (Song et al. 2026). The appeal of the third approach is that a physically grounded invariant does not depend on which attacks were recorded.

This line of work addresses distribution shift, in which the test attacks are drawn from a different distribution than the training attacks but are still drawn rather than chosen. Adversarial selection is a distinct axis: the attack is not sampled from any distribution but optimised against the deployed system. The two questions are complementary, and a library of transmitted attacks of the kind constructed here provides material against which invariant based detectors can subsequently be examined.

2.4 Adversarial robustness of learned classifiers

Classifiers fitted to data can be driven across their own decision boundaries by small, deliberately chosen input perturbations, an observation that originated in image classification and has since been shown to be general (Szegedy et al. 2014; Goodfellow et al. 2015). The associated literature covers attack construction under model access (Madry et al. 2018), methodological standards for evaluating defensive claims (Carlini et al. 2019), and the framing of the problem as a security question (Biggio and Roli 2018).

Two results from this literature bear directly on the evaluation of spoofing detectors. The first concerns adversary knowledge. Attacks that differentiate the loss require access to model parameters, whereas decision based attacks require only the classifier’s output label and are nonetheless effective (Brendel et al. 2018; Chen et al. 2020). A model that withstands a gradient attack has therefore not been shown robust, because a model that supplies uninformative gradients may still be crossed by a search that queries only decisions, a failure mode termed gradient masking (Athalye et al. 2018). The second concerns cost. Apparent robustness obtained by displacing a decision threshold is not a gain, since any classifier can be made difficult to cross by flagging nearly all inputs; the accuracy and robustness of a fitted classifier trade against one another (Tsipras et al. 2019). A robustness claim is therefore interpretable only when reported jointly with the false alarm rate that purchased it, and the present study adopts that convention throughout.

2.5 Physical realizability and the evaluation gap

Transferring these results to a sensing system introduces a requirement absent from the image setting. A perturbation expressed as a displacement of a feature vector corresponds to an attack only if some physical action on the sensor produces that displacement. Research on other sensing modalities has addressed this by constructing attacks in the physical domain, including against automotive ranging sensors (Petit et al. 2015; Cao et al. 2021), against speech recognition (Carlini and Wagner 2018) and against synthetic aperture radar imagery (Lin et al. 2023). In the GNSS setting, adversarial perturbation of detector inputs has been reported (An et al. 2025).

For a GNSS receiver the requirement is stringent, because the observables a detector reads are not independent coordinates. Several are computed from the same correlator accumulations within a single integration period, so displacing one while holding the others fixed describes a receiver state that no incident waveform produces. Establishing that a given feature space perturbation is realizable therefore requires an argument about receiver processing that is specific to the observable set in use. Evaluating at the waveform level removes the need for such an argument by construction, since the adversary acts on transmitter parameters and the observables are whatever the receiver computes while tracking the resulting signal. This is the approach adopted in the present study, and Section 6.5 reports the corresponding feature space evaluation alongside it so that the two can be compared directly.

3 Receiver Observables and Signal Tracking

The detectors evaluated here operate on observables a receiver computes while tracking a signal. This section specifies the processing that produces them, the observable set retained, the criterion by which that set was selected, and the labelling procedure, so that the effect of a transmitted attack on each quantity can be traced through the chain.

3.1 Receiver processing

A GPS L1 C/A satellite transmits a carrier modulated by a pseudorandom spreading code of 1023 chips repeating every millisecond, and by navigation data at 50 bits per second. The code is public and distinct for each satellite, which is what permits a counterfeit to be generated at all.

Reception proceeds in two stages. In acquisition the receiver searches jointly over code phase and carrier frequency until it finds the alignment at which its locally generated replica correlates strongly with the incoming signal. In tracking it maintains that alignment as the satellite geometry and the receiver clock evolve. Tracking is implemented by three feedback loops operating concurrently. The delay locked loop maintains code alignment by correlating the incoming signal against early, prompt and late replicas and steering so that the early and late correlations remain equal, which holds the prompt replica on the correlation peak. The phase locked loop maintains carrier phase alignment. The frequency locked loop performs the equivalent function

Table 1 The nine measurements each detector reads, per satellite per epoch.

Measurement	Units	What it is
C/N_0	dB-Hz	Received signal strength relative to the noise density
Mean C/N_0	dB-Hz	The same, averaged over the preceding second
Noise estimate	–	Correlator output away from the peak, used as the noise reference
Doppler	Hz	Frequency offset of the carrier from its nominal value
Prompt in phase	–	Correlation with the on peak code copy, in phase component
Prompt quadrature	–	The same, quadrature component
Delay lock discriminator	chips	Difference between the early and late correlations, the code tracking error
Phase lock indicator	–	How well the phase locked loop is holding
Frequency lock indicator	–	How well the frequency locked loop is holding

for carrier frequency and is more tolerant of low signal to noise ratio, so it assists when phase lock is marginal.

The loops produce a set of quantities at each integration interval. These quantities are the detector inputs used throughout this study. They are not selected for convenience of classification: they are what the receiver computes in order to function, and a detector operating inside a receiver would have access to precisely this set.

3.2 Observable measurements

Table 1 lists the nine observables, exported once per epoch for each tracked satellite.

The set spans the three aspects of tracking that a spoofing attack perturbs. Received power is represented by the first three entries, which respond to the counterfeit’s transmitted power relative to the authentic signal. Carrier behaviour is represented by the Doppler, the two prompt correlator components and the phase and frequency lock indicators, which respond to the counterfeit’s carrier frequency and phase. Code behaviour is represented by the delay lock discriminator, which responds to the displacement between the counterfeit code and the authentic code and therefore to the pull-off rate. Figure 3 shows the marginal distribution of each observable under the two classes; the classes overlap in every single observable, which is why a learned detector that combines them can outperform any univariate threshold.

3.3 Selection of the observable set

A software receiver exposes more quantities than the nine retained, and several of the additional ones are exact deterministic functions of quantities already in the set. The prompt correlator power is the sum of squares of the two prompt components; the

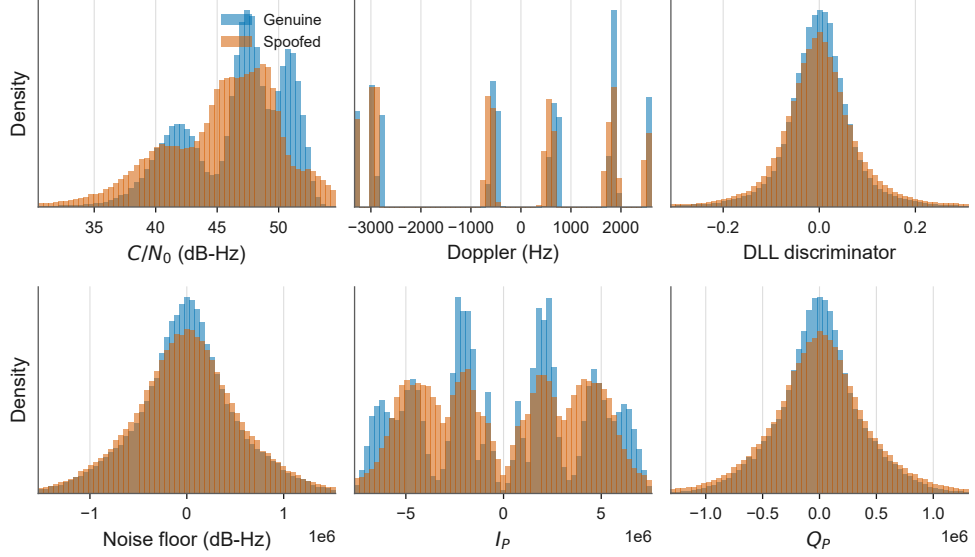


Fig. 3 Distribution of each of the nine measurements for authentic and for spoofed epochs. The classes overlap in every measurement, which is why a detector combines them instead of thresholding any one.

prompt phase is the arctangent of their ratio; the carrier frequency is the Doppler displaced by a fixed nominal value.

Such derived quantities are excluded, for two reasons. They contribute no information beyond their parents, so their inclusion cannot improve an optimal classifier. More importantly, their presence would admit an evaluation artefact: a perturbation that displaced a derived quantity without correspondingly displacing its parents specifies a receiver state that no incident waveform can produce, and any attack exploiting such a displacement would be an artefact of the feature representation. The nine retained observables are functionally independent, in the sense that none is an exact function of the others, which eliminates that particular failure mode.

Functional independence is a property of the representation and does not by itself establish that an arbitrary configuration of the nine is physically attainable, since independence among coordinates says nothing about the image of the map from transmitter parameters to observables. In the waveform level evaluation the question is not reached, because the observables are by construction whatever the receiver computes while tracking a transmitted signal. It is reached in the feature space comparison of Section 6.5, and is treated there. Figure 4 shows the joint evolution of the observables for one satellite across the onset of a recorded attack.

3.4 Labelling and data partitioning

Each epoch is labelled according to whether the signal being tracked at that instant was authentic or contained a counterfeit, following the documented onset time within each recording. Epochs spanning the takeover transient are discarded, because during

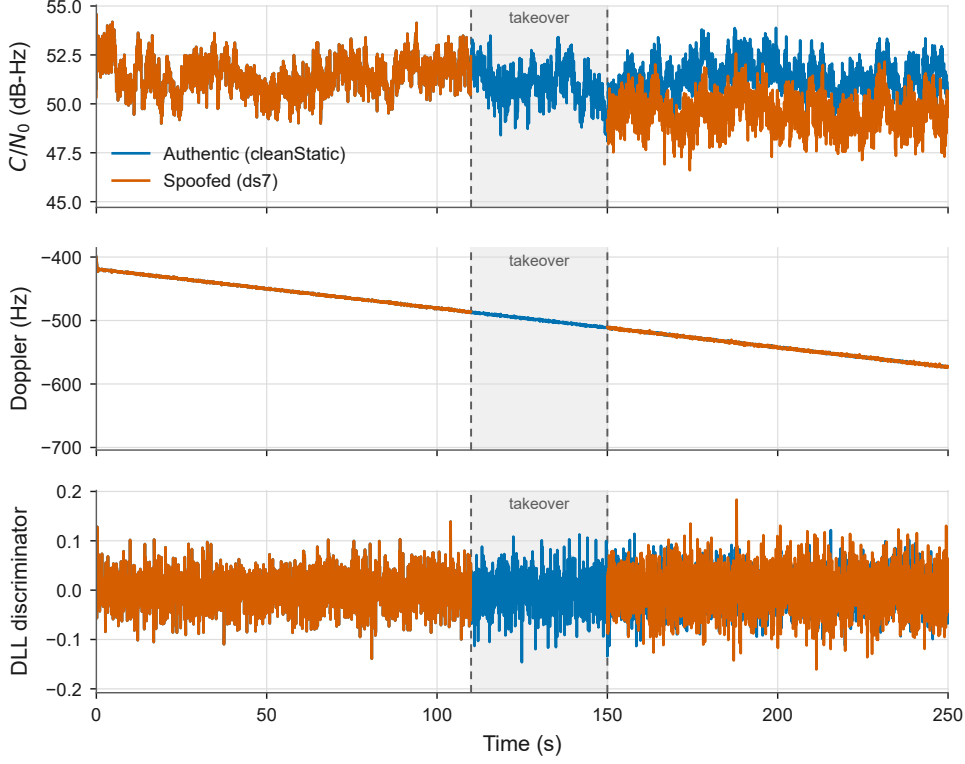


Fig. 4 Time history of the measurements for one satellite, PRN 23, across the onset of a recorded attack. The measurements move together as the counterfeit takes over the tracking loops, which is the signature a detector reads.

handover the tracked signal is neither fully authentic nor fully under the adversary’s control and the label would be ambiguous.

Both classes are drawn from within the same recording. This is a deliberate constraint on corpus construction. Were authentic epochs taken from one recording and spoofed epochs from another, a classifier could attain high apparent accuracy by identifying the recording of origin through front end, antenna or clock characteristics, which is not spoofing detection and would not transfer to deployment. Drawing both classes from a single recording, with a common front end and a common clock, removes that confound.

Training, validation and test partitions are taken as disjoint contiguous blocks in time, with a margin discarded at each boundary. Adjacent epochs of the same satellite are strongly correlated, since the observables evolve continuously and the integration interval is short relative to the dynamics, so a randomised partition would place near duplicates of the same observation on both sides of the split and inflate the measured accuracy.

Table 2 The attacker’s controls. Each row is a transmitter setting, not a property of the victim’s measurements. One choice of all five defines one attack.

Setting	Symbol	What it does
Power advantage	α	Strength of the counterfeit relative to the authentic signal, in decibels
Code offset	τ_0	Alignment of the counterfeit code at the start, in chips
Code pull-off rate	$\dot{\tau}$	Rate the counterfeit code is dragged away from the authentic code
Carrier offset	Δf	Frequency of the counterfeit carrier relative to the authentic carrier
Carrier phase	ϕ_0	Phase of the counterfeit carrier relative to the authentic carrier

4 Threat Model and Attack Space

This section defines the adversary, the parameters under its control, the conditions its transmission must satisfy to capture a receiver, and the knowledge it is assumed to possess. Every adversary action in this study is a choice of transmitter settings, so the threat model is stated in terms of a transmitter rather than in terms of a displacement in a feature space.

4.1 Adversary model and objective

The adversary is a spoofer: a transmitter emitting counterfeit GPS L1 C/A signals towards a victim receiver. Its objective is to cause the victim to report a position or a time of the adversary’s choosing while the victim continues to operate nominally, producing a converged navigation solution with residuals of ordinary magnitude.

The adversary’s only channel to the victim is the radio interface. It does not modify the victim’s software, read its internal state, or alter the observables the victim computes. Its influence on those observables is mediated entirely by the receiver’s own acquisition and tracking of the composite signal, which is what determines the adversary’s variables in this study.

4.2 Transmitter controlled parameters

Table 2 lists the parameters the adversary sets. A choice of all five defines one attack.

Two of these parameters carry the attack. The power advantage α determines whether the victim’s tracking loops transfer to the counterfeit at all, and the pull-off rate $\dot{\tau}$ determines how rapidly the victim’s reported range moves once transfer has occurred. One chip of GPS L1 C/A code corresponds to $c/f_{\text{chip}} = 293.05$ m of range, where c is the speed of light and $f_{\text{chip}} = 1.023$ MHz is the chipping rate, so a pull-off rate expressed in chips per second converts directly to metres per second of induced range error. The remaining three parameters govern the alignment of the counterfeit at onset, and therefore govern whether the capture conditions of Section 4.3 are met.

4.3 Receiver takeover conditions

The adversary cannot select these parameters arbitrarily and still capture a receiver that is already tracking the authentic signals. Two conditions govern capture (Psiaki and Humphreys 2016; Kerns et al. 2014).

The counterfeit must arrive aligned with the authentic signal in code phase, carrier phase and Doppler frequency, so that the victim’s correlator observes a single undistorted peak rather than two competing peaks. A misaligned counterfeit does not transfer the tracking loops; it perturbs the victim’s observables without displacing its position or time solution, and therefore constitutes interference rather than spoofing. The counterfeit must additionally exceed the authentic signal in received power, since code tracking follows the dominant correlation peak. In practice the power advantage employed is small, typically a few decibels, because a large advantage is itself observable in the receiver’s automatic gain control and in the reported C/N_0 .

Only once both conditions hold does the adversary begin to displace the code. These conditions define the subset of the parameter space in which an attack is operative, and Section 5 applies them to partition the swept library into transmissions that capture the receiver and transmissions that do not. Detector performance is reported on the former, so that no detector is credited for flagging a transmission that would not have succeeded.

4.4 Knowledge and access assumptions

The adversary is assumed to have no knowledge of the victim’s detector. It does not know which of the fourteen detectors is deployed, what data that detector was trained on, what decision threshold it uses, or what score it assigns to any observation. No gradient of any detector is computed and no detector is queried at any point in the attack construction.

In the taxonomy of Section 2.4 this is weaker than a decision based adversary, which at least observes the classifier’s output label. It corresponds to a spoofer transmitting without feedback from the victim, which is the operationally realistic case for an attack on a receiver whose alarm state is not externally visible. The detection failures reported in Section 6 are therefore failures against an adversary with no information about the system it defeats.

4.5 Scope of the adversary model

Four further restrictions delimit the adversary modelled here relative to the capability a spoofer may possess in general.

- **No detector knowledge.** As stated above, the transmission is not shaped against any specific model.
- **All tracked satellites are spoofed.** An adversary that leaves one satellite authentic gains an additional degree of freedom, at the cost of an inconsistency between that satellite and the remainder which a cross satellite check may identify.

- **A single signal is transmitted**, GPS L1 C/A. A receiver combining multiple constellations or multiple frequencies retains evidence that this adversary does not attempt to defeat.
- **The victim is stationary and its position is surveyed**, so a common pull-off applied to all channels is absorbed into the receiver clock estimate and manifests as a timing error.

These restrictions determine how the results should be interpreted. Each detection failure reported in Section 6 is obtained against this restricted adversary, so the measured vulnerability is a lower bound on the vulnerability to an adversary that relaxes any of them. The fourth restriction additionally fixes the observable consequence measured in Section 6.4: because a common pull-off on all channels is absorbed by the clock, the induced error appears in the timing solution, and it is that error which is compared against the commanded rate.

5 Experimental Methodology

This section describes how the counterfeit signal is generated, how it is checked against a recorded attack, how the library of transmitter settings is built, what the fourteen detectors are, and how they are set to a common operating point.

5.1 Corpus and receiver configuration

The authentic signal is the cleanStatic recording of the Texas Spoofing Test Battery (Humphreys et al. 2012). It holds raw in-phase and quadrature samples of the GPS L1 band taken at a surveyed stationary antenna, at 25 million complex samples per second with 16 bits in each component. It contains no spoofing, so it supplies an authentic signal that a counterfeit can be added to.

All tracking is performed by FGI-GSRx (Finnish Geospatial Research Institute 2022), an open software receiver. Using a published receiver matters for two reasons. The measurements the detectors read are the ones a real integrity monitor would compute, and any reader can reproduce them from the same recordings with the same software.

5.2 Spoofing waveform generation

The counterfeit is built as radio samples and then added to the authentic samples.

The generator is seeded from the receiver’s own tracking of the authentic recording. For each satellite the receiver reports, at every millisecond, where it found the code, what carrier frequency it is tracking and how strong the signal is. Those quantities describe the authentic signal as the receiver sees it, so they are exactly what a counterfeit must match in order to arrive aligned. The generator reads them and produces, for each satellite, a replica of the GPS L1 C/A signal at the same code phase, the same carrier frequency and the same amplitude.

The attacker’s settings of Table 2 are then applied to that replica. The power advantage scales its amplitude. The code offset and pull-off rate displace its code,

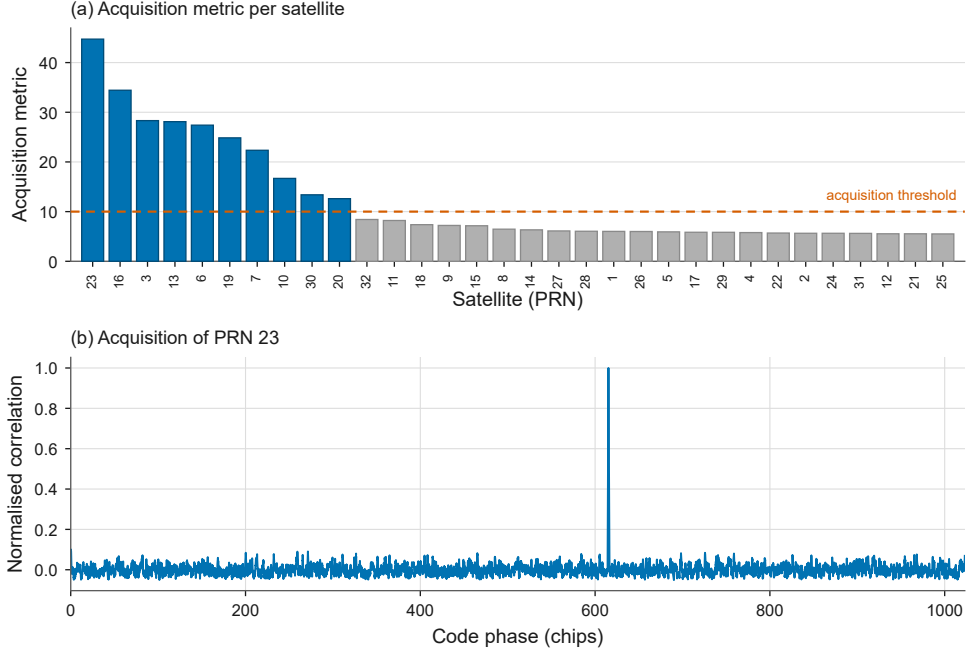


Fig. 5 Validation of the software receiver on the authentic recording. Panel (a) shows the acquisition metric for each satellite, and panel (b) shows the correlation against code phase for one satellite. Both match what a correctly operating receiver produces, which establishes that the measurements used throughout come from sound tracking.

with the displacement growing at the chosen rate. The carrier offset and phase shift its carrier.

The counterfeit and the authentic samples are summed, and the result passes through a model of the receiver front end before being written. The front end applies automatic gain control, which is the mechanism a real receiver uses to keep the signal inside the range its converter can represent. When added power would drive the sum beyond that range, the gain is reduced so that the sum still fits. This detail matters. Without it, added power would clip, and the clipping would be an artefact of the file format instead of a property of the attack.

The written file is then tracked by FGI-GSRx exactly as the authentic recording is, and the measurements it produces are what the detectors read.

Two properties follow from building the attack this way. Every constraint a receiver imposes is satisfied because a receiver produced the measurements, which covers the continuity of code and carrier, the bandwidth and settling of the tracking loops, the behaviour of the automatic gain control and the smoothness of the measurements from one epoch to the next. No projection onto a set of allowed values is applied at any point, because there is no perturbation to project. Figure 5 shows the checks on the receiver itself.

5.3 Generator validation

A generator that cannot reproduce a documented attack is not evidence about attacks that were never recorded. The check uses data set ds7 of the battery, whose construction is published in full (Humphreys 2016).

That document specifies ds7 as a power matched time push carried out with carrier phase alignment. During the takeover the counterfeit phasor is $P_s(t) = A_s(t)e^{j\theta(t)}$, with the angle θ rising from $\pi/2$ to π and the amplitude following $A_s(t) = -2A_a \cos \theta(t)$, where A_a is the authentic amplitude. Those two choices together give a combined phasor

$$P_a + P_s = A_a + A_s e^{j\theta} = -A_a e^{j2\theta}, \quad (1)$$

whose magnitude stays at A_a while its angle turns from 0 to π . The victim's loops are therefore walked onto the counterfeit with no step in power and no step in phase. After the takeover the counterfeit holds at twice the authentic amplitude in antipodal phase, so the sum returns to magnitude A_a , and the document records that the victim's measurements match the clean recording apart from a carrier phase that is greater by π . From then the code is dragged at 1.2 metres per second while the counterfeit Doppler is held equal to the authentic Doppler.

Running the generator at those settings reproduces the recorded measurements. Figure 6 reports the comparison. Seven of the eight satellites agree within half a decibel of carrier-to-noise density ratio. The eighth differs by more, and the difference is not isolated: across the satellites the gap between generated and recorded grows with the strength of the satellite, with a Spearman rank correlation of +0.905 at $p = 0.002$, and the trend remains when the strongest satellite is removed. The cause is that the generated counterfeit is an amplitude matched replica, so the cancellation in Equation 1 is cleaner than the recorded spoofer's hardware achieved, which leaves the generated attack slightly louder. A slightly louder attack is easier to detect, so this difference understates what an attacker can do.

5.4 Transmitted attack library

The attacker's choice set is expressed as a library. Each entry is one combination of the settings in Table 2, and for each the generator produces a composite file which the receiver tracks and the exporter turns into measurements. The library holds 209 entries.

Two families of setting are covered. In the first the counterfeit carrier is in phase with the authentic carrier, and the power advantage is swept from -3 to 10 decibels. The second follows the law of Equation 1: for a counterfeit of amplitude ratio g at phase φ , the received power is unchanged when

$$g = -2 \cos \varphi, \quad (2)$$

so a counterfeit on this locus is invisible to a test that watches received power alone. The ds7 attack sits on it at $\varphi = \pi$ and $g = 2$. Both families are crossed with seven code pull-off rates from zero to 147 metres per second and three carrier alignment settings.

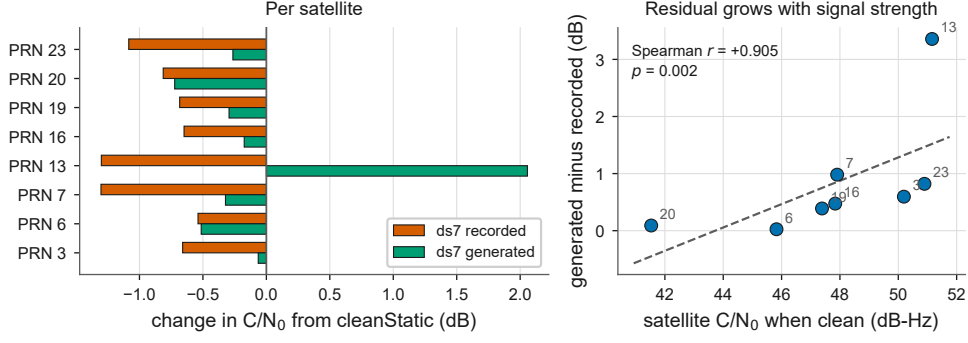


Fig. 6 The generator reproduces the recorded ds7 attack. Left, the change in carrier-to-noise density ratio from the clean recording, for the recorded attack and for the generated one, per satellite. Right, the residual between them grows with satellite strength, so the strongest satellite is the extreme of a trend and not an isolated disagreement. The generated attack sits slightly above the recorded one, which makes it easier to detect.

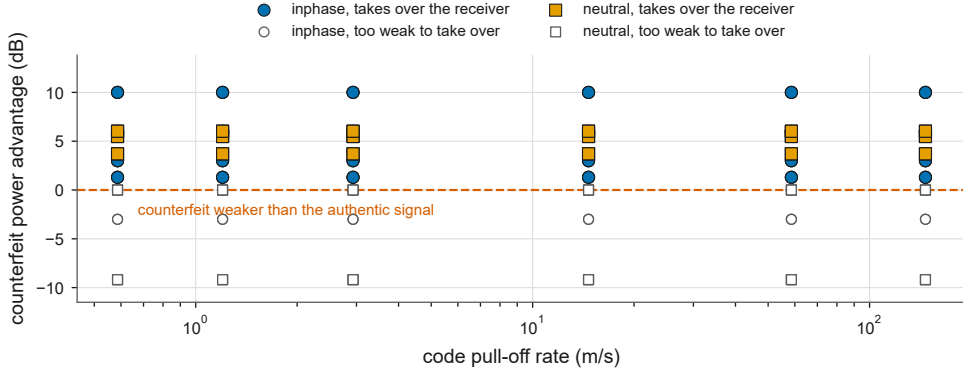


Fig. 7 The transmitter settings in the library. Filled markers are the 126 settings where the counterfeit is stronger than the authentic signal and is dragging the code, which are the conditions for taking over a receiver. Open markers are settings that do not meet them. Circles are the in phase family and squares follow the power neutral law of Equation 2.

Section 4 gives the two conditions a counterfeit must satisfy to take over a receiver. Applied to the library, they select the entries where the counterfeit is stronger than the authentic signal and is dragging the code. That is 126 of the 209 entries, and those 126 are the attacks the detectors are measured against. Figure 7 shows the library and the subset.

5.5 Detector models and training

Fourteen detectors are tested. They are listed in Table 3 with what each one does, so that the comparison can be read without reference to the machine learning literature. All fourteen read the same nine measurements described in Section 3.

Table 3 The fourteen detectors and the decision rule each one applies. The classical models decide from the measurements of a single epoch. The deep networks decide from a window of consecutive epochs, so they can use how the measurements change.

Detector	Family	How it decides
Decision tree	classical	A sequence of threshold tests on single measurements, forming one branching rule (Breiman 2001)
Random forest	classical	Many decision trees fitted to different samples of the data, whose votes are averaged (Breiman 2001)
Gradient boosting	classical	Trees added one at a time, each fitted to the errors left by the previous ones (Friedman 2001)
XGBoost	classical	Gradient boosting with additional penalties on tree complexity (Chen and Guestrin 2016)
LightGBM	classical	Gradient boosting that grows trees leaf by leaf for speed on large data (Ke et al. 2017)
Nearest neighbour	classical	Labels an epoch by the labels of the most similar training epochs
Support vector machine	classical	Places a boundary separating the classes with the widest margin, using a radial basis kernel
Multilayer perceptron	classical	A small feed forward network of fully connected layers
One dimensional convolutional network	deep	Filters slid along the window respond to local patterns in time
Long short term memory	deep	Carries a state along the window, so earlier epochs inform later ones (Hochreiter and Schmidhuber 1997)
Bidirectional long short term memory	deep	The same, traversing the window in both directions
Convolutional recurrent hybrid	deep	Convolutional filters feeding a recurrent layer
Transformer	deep	Weighs every epoch in the window against every other through attention (Vaswani et al. 2017)
Temporal convolutional network	deep	Dilated convolutions reaching across the window without recurrence (Bai et al. 2018)

The set spans the families used in the detection literature reviewed in Section 2.2, covering tree ensembles, instance based classification, kernel methods, a feed forward network and the convolutional, recurrent, hybrid and attention branches of sequence modelling.

5.6 Feature space comparison

The attacks described above are transmitted. To show what a study that perturbed the measurements instead would have concluded, each detector is also measured a second way, and that measurement is reported alongside as a comparison.

The comparison uses a decision based boundary attack. It is given only the detector’s output decision for a sample, authentic or spoofed, and no gradient, no score and no knowledge of the model. Starting from a spoofed sample the detector flags, and from an authentic sample it does not, the attack performs a binary search along the line between them to locate the point where the decision changes, then reduces the largest single change across the nine measurements while keeping the decision on the authentic side. The quantity reported is the smallest such change, measured in the normalised space where each measurement is scaled to the unit interval, so the nine are comparable.

This is the appropriate comparison to make. A gradient attack would need access to the detector’s internals, and Section 4 states that the attacker in this study has none. A decision based attack needs only the output decision, so it is the strongest attack available to an adversary that cannot see inside the model (Brendel et al. 2018; Chen et al. 2020). It is also the attack that a non-differentiable detector cannot evade by denying gradients (Athalye et al. 2018).

The comparison is reported for one purpose. It shows where a measurement space evaluation and a transmitted attack agree about a detector and where they do not, which Section 6.5 sets out. It is not used to support any claim about what a spoofer can achieve, because a perturbation of the measurements is not by itself a signal a transmitter can send.

5.7 Operating point calibration

A detector reports a score, and a threshold turns that score into a decision. Moving the threshold trades two quantities against each other. Recall is the fraction of spoofed samples that are flagged. The false alarm rate is the fraction of authentic samples that are flagged. Raising the threshold lowers both, and lowering it raises both, so two detectors read at different thresholds cannot be compared.

Every detector in this study is therefore set to one common operating point. The threshold is the highest value at which the detector still flags 95 percent of the spoofed samples in clean validation data,

$$\eta^* = \max\{\eta : R_{\text{val}}(\eta) \geq 0.95\}, \quad (3)$$

where R_{val} is recall on that data. A recall floor is the safety first choice for an integrity monitor, because missing an attack is the failure that matters. The false alarm rate each detector pays to meet that floor is reported alongside every result, since a detector can always meet a recall floor by flagging everything.

The data are divided so that training, validation and test come from separate blocks of each recording, with a gap discarded at every block boundary. Adjacent epochs of the same satellite are nearly identical, because the tracking loops have time constants of order a second, so a division that scattered neighbouring epochs across the blocks

Table 4 Every detector at the common operating point of Equation 3, ordered by false alarm rate. Passed counts the transmitted take over attacks, out of 126, for which the detector flags fewer than half the spoofed samples. Lowest detection is the smallest fraction of an attack’s samples the detector flagged. The final column is the smallest perturbation of the measurements alone that crosses the decision boundary, for comparison.

Detector	Family	Clean data		Transmitted attacks		Perturbation
		F_1	False alarm	Passed	Lowest det.	L_∞
Random forest	classical	0.833	0.337	55 (43.7%)	0.001	0.025
Gradient boosting	classical	0.827	0.350	27 (21.4%)	0.166	0.019
Temporal convolutional	deep	0.805	0.427	4 (3.2%)	0.469	0.041
XGBoost	classical	0.811	0.451	28 (22.2%)	0.088	0.020
Nearest neighbour	classical	0.800	0.462	0 (0.0%)	0.692	0.055
LightGBM	classical	0.804	0.464	27 (21.4%)	0.151	0.021
Decision tree	classical	0.787	0.522	21 (16.7%)	0.139	0.023
Multilayer perceptron	classical	0.758	0.587	14 (11.1%)	0.171	0.039
Bidirectional LSTM	deep	0.741	0.662	0 (0.0%)	0.712	0.064
Long short term memory	deep	0.735	0.679	0 (0.0%)	0.568	0.065
Convolutional recurrent	deep	0.734	0.693	0 (0.0%)	0.725	0.067
Convolutional network	deep	0.725	0.737	0 (0.0%)	0.636	0.058
Transformer	deep	0.707	0.794	18 (14.3%)	0.004	0.058
Support vector machine	classical	0.682	0.960	53 (42.1%)	0.074	0.111

would place near copies of the same measurement in training and in test and would inflate every score.

6 Results

Results are reported in four parts. The first is how the detectors perform on clean data, which fixes the baseline. The second is how they perform against the transmitted attacks. The third follows one missed attack into the navigation solution. The fourth compares these measurements against what a perturbation of the measurements alone would have suggested.

6.1 Clean performance and false alarm cost

Every detector is set to the common operating point of Equation 3, which is the highest threshold at which it still flags 95 percent of the spoofed samples in clean validation data. Table 4 reports what each detector achieves there.

Two quantities describe clean performance. The F_1 score is the harmonic mean of precision and recall, so it summarises detection quality in one number. The false alarm rate is the fraction of authentic samples the detector flags, and it is the cost of meeting the recall floor.

The random forest achieves the highest F_1 score at 0.833 and the lowest false alarm rate at 0.337. Gradient boosting follows at 0.827 and 0.350. The support vector machine is last on both counts, at 0.682 and 0.960.

The false alarm rates deserve attention on their own. Across the fourteen detectors they run from 0.337 to 0.960. Even the best of them flags about a third of authentic samples as spoofed. On this evidence no detector in the study is usable as a standalone single epoch integrity monitor, before any attacker is considered. A fielded monitor would combine decisions over several epochs, trading detection latency for a lower

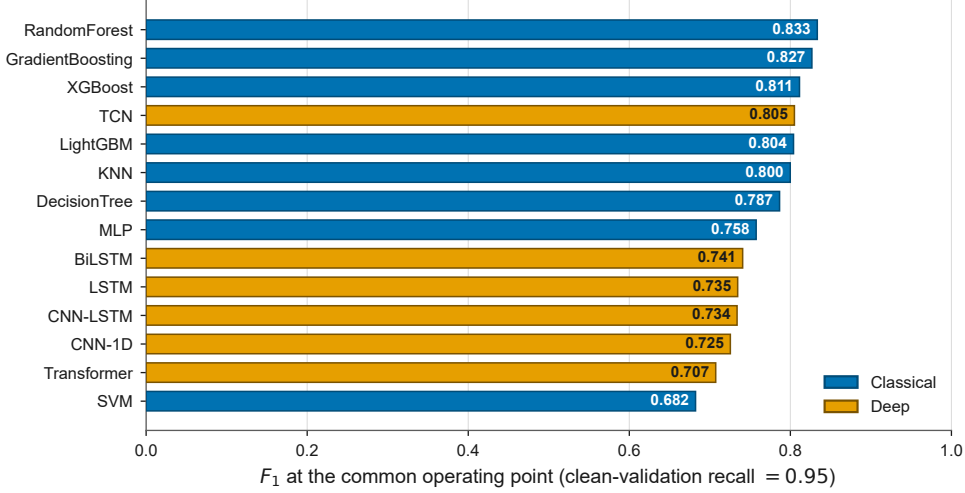


Fig. 8 Clean F_1 for every detector at the common operating point, the threshold at which each flags 95 percent of spoofed samples in clean validation data. Classical and deep detectors are interleaved, so no family dominates before an attacker is considered.

rate of false alarms, and the per epoch numbers reported here are the input to such a scheme. Figure 8 shows the clean scores at that operating point.

6.2 Detector performance under transmitted attacks

Each detector is then run against the 126 transmitted settings that meet the conditions for taking over a receiver. For each setting the detection rate is the fraction of that attack’s spoofed samples the detector flags. Because every sample in these files is spoofed, the detection rate is the detector’s recall against that particular attack. A setting counts as passed when the detector flags fewer than half of its samples.

Table 4 and Figure 9 report the outcome.

The detector that performs best on clean data is the one most often passed. The random forest fails to flag 55 of the 126 attacks, which is 43.7 percent of them. Against one setting it flags 0.1 percent of the spoofed samples, so the attack is present throughout and the detector reports authentic almost everywhere.

Gradient boosting fails to flag 27, XGBoost 28 and LightGBM 27. The support vector machine fails to flag 53, and it does so while flagging 96 percent of authentic samples, so it is neither accurate nor resistant.

Five detectors flag every one of the 126 attacks. They are the nearest neighbour classifier, the one dimensional convolutional network, the convolutional recurrent hybrid, the bidirectional long short term memory and the long short term memory. Their lowest detection rate against any attack stays between 0.568 and 0.725. Their false alarm rates on authentic data run from 0.462 to 0.737, so between 46 and 74 percent of authentic samples are flagged as attacks. Their resistance is bought by flagging a large part of the authentic data, which is the trade described in Section 2.4 and not a property that could be used in a receiver.

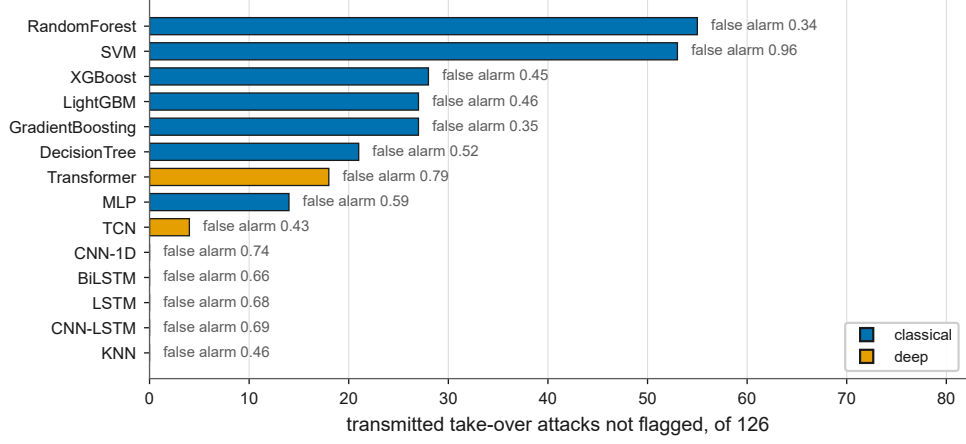


Fig. 9 Transmitted take over attacks each detector fails to flag, out of 126. The clean false alarm rate each detector pays at the common operating point is printed beside its bar. The detector with the lowest false alarm rate fails to flag the most attacks.

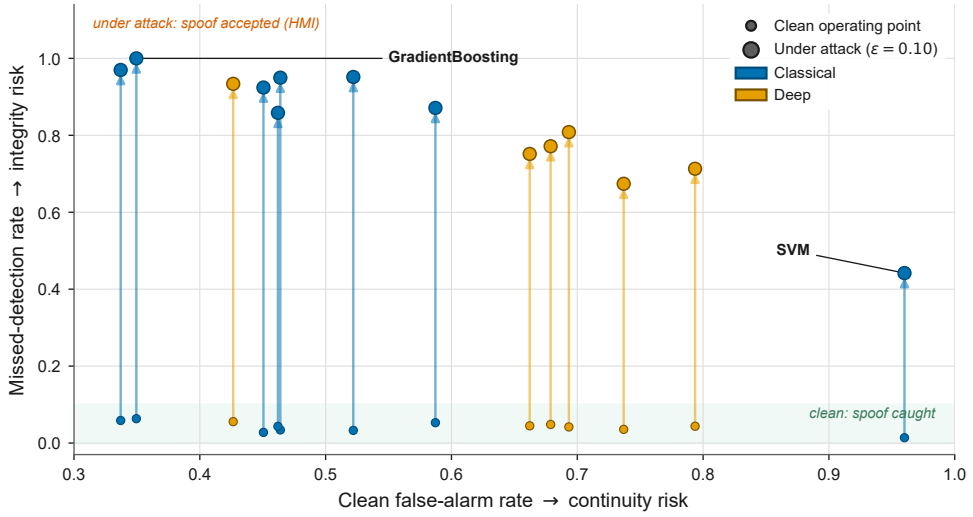


Fig. 10 Each detector placed by the two risks a navigation integrity monitor trades. The horizontal axis is the false alarm rate on authentic data, which drives continuity risk. The vertical axis is the missed detection rate, which drives integrity risk. No detector reaches the lower left corner.

Taken together, no detector in the study is both usable and resistant. The two with the lowest false alarm rates are passed by 43.7 and 21.4 percent of the attacks. The five that are never passed carry false alarm rates that rule them out. Figure 10 places the detectors on the two axes an integrity monitor trades.

6.3 A single transmitter setting evading six detectors

The counts above describe an attacker that transmits every setting in turn. A real attacker chooses one setting and transmits it. The relevant question is therefore what a single choice achieves.

The best single setting is a counterfeit three decibels stronger than the authentic signal, in phase with it, dragging the code at 14.65 metres per second. Transmitted once, it is passed by six of the fourteen detectors at the same time. Those six are the decision tree, the random forest, gradient boosting, XGBoost, LightGBM and the Transformer, so every tree based detector in the study is among them, including the one with the best clean performance, which flags 1.2 percent of the samples.

This is the same setting that Section 6.4 follows through the navigation stage. One transmitted configuration therefore passes six detectors and drives the receiver clock at the commanded rate. The attacker needs no knowledge of which detector is running, because this single choice covers six of them.

A second setting, identical except that the code is dragged at 2.93 metres per second, passes the same six detectors. The attacker can therefore trade the speed of the induced error against nothing at all, since both rates pass the same set.

6.4 Navigation domain consequence

Detection failure matters only if the attack it misses does something. One of the missed settings is therefore followed through the navigation stage, which is the part of the receiver that turns the measurements into a position and a clock estimate.

The setting is a counterfeit three decibels stronger than the authentic signal, dragging the code at 14.65 metres per second. The random forest flags 1.2 percent of its samples. The attack drags every satellite by the same amount, and a stationary receiver absorbs a common change in range into its clock estimate, so the expected effect is a drift of the receiver clock at

$$\frac{\dot{\tau} c / f_{\text{chip}}}{c} = \frac{14.65}{c} = 48.87 \text{ ns/s.} \quad (4)$$

The receiver's own clock drift is measured first, by running the same window with the counterfeit switched off. It is 0.12 nanoseconds per second, so it is negligible against the rates under test, and it is subtracted from every figure below.

The corrected drift is 48.89 nanoseconds per second against a commanded 48.87, a difference of 0.02 nanoseconds per second, or 0.05 percent. Figure 11 shows the clock estimate over the window together with the commanded rate. The navigation solution converged at every epoch with a figure of merit below one metre, so the receiver was producing an ordinary fix while its clock was being driven.

The right panel of Figure 11 shows the ds7 takeover at 1.2 metres per second. This attack carries a takeover of 20 seconds before its code begins to drift, and the tracking loops need time to settle once they have been captured. Fitting across the whole window therefore mixes the settling transient with the drift. The measured slope over successive 20 second segments is -6.27 , -7.50 , -4.72 and -4.68 nanoseconds per second, so the loops have settled about 30 seconds in.

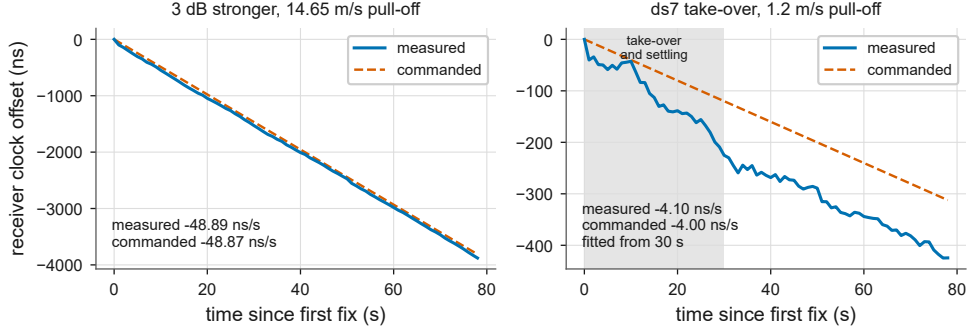


Fig. 11 The attack reaches the navigation solution. Receiver clock offset over the window for two transmitted attacks, with the rate each transmitter was set to impose. In the left panel the measured drift is within 0.05 percent of the commanded rate. The detector with the best clean performance reports 98.8 percent of the samples in that attack as authentic.

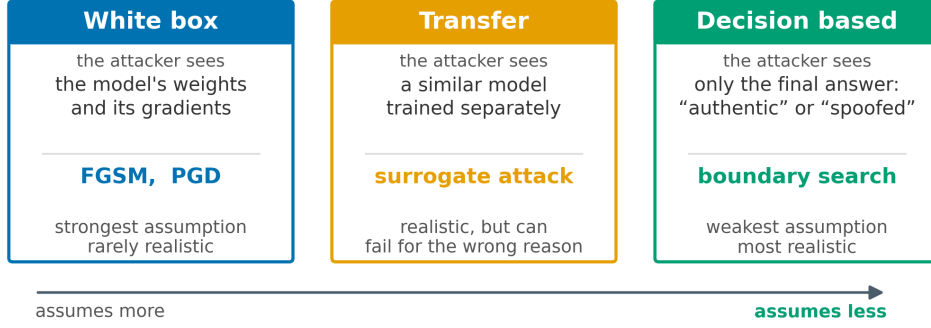


Fig. 12 What an attacker is assumed to know. A white box attacker holds the model and its gradients, a transfer attacker holds a separately trained substitute, and a decision based attacker sees only the output decision. The comparison in this section uses the last of these, which assumes least.

Fitted after settling, the corrected drift is 4.10 nanoseconds per second against a commanded 4.00, a difference of 2.3 percent. Both attacks therefore drive the receiver clock at the rate the transmitter was set to impose, one of them through a takeover and one without. Figure 12 sets out what each kind of attack is assumed to know.

6.5 Comparison with feature space evaluation

The same fourteen detectors were also measured with the decision based boundary attack of Section 5.6, which perturbs the measurements directly without generating a signal. Table 4 reports the size of the smallest perturbation that crosses each detector's decision boundary, expressed in the normalised measurement space.

The two views agree on which detectors are fragile. The tree based detectors need the smallest perturbations, from 0.019 to 0.025, and they are also the detectors most

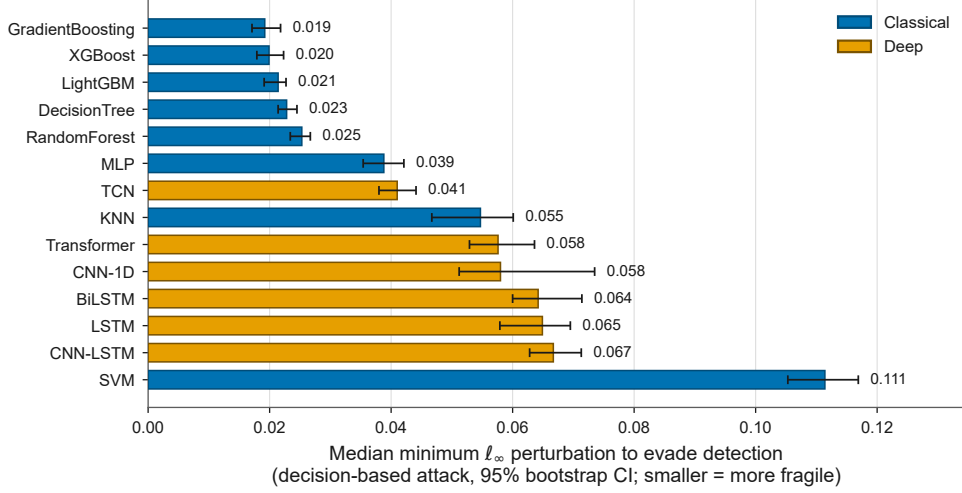


Fig. 13 Median smallest perturbation of the measurements alone that moves each detector across its decision boundary, under the decision based attack. Larger is more resistant. This is the comparison measurement, not the transmitted attack.

often passed by a transmitted attack. The agreement is useful, because it shows the fragility is a property of those detectors and not an artefact of either measurement.

The two views disagree on the support vector machine. It needs by far the largest perturbation at 0.111, which taken alone would read as the most robust detector in the study. Its false alarm rate of 0.960 explains this. A detector that flags almost everything is hard to push into flagging, and the transmitted measurement shows the same detector being passed by 53 of the 126 attacks. Reading the perturbation size without the false alarm rate would have inverted the conclusion. Figure 13 gives the resulting ranking.

Figure 14 compares that ranking against the clean one.

7 Discussion

7.1 Principal findings

Three findings follow from Section 6, each obtained against attacks that were synthesised as waveforms and tracked through the receiver.

Clean detection performance does not predict resistance to transmitted attack, and at the top of the ranking the two orderings are close to inverted. The random forest attains both the highest clean F_1 score and the lowest false alarm rate of the fourteen detectors, and it is nonetheless evaded by 43.7 percent of the transmitted takeover attacks. This is not a marginal effect confined to one model: the four best detectors on clean data occupy four of the five worst positions under transmitted attack. Because all fourteen detectors are read at the same recall floor, the inversion cannot be attributed to differences in threshold placement, and because the attacks were selected without

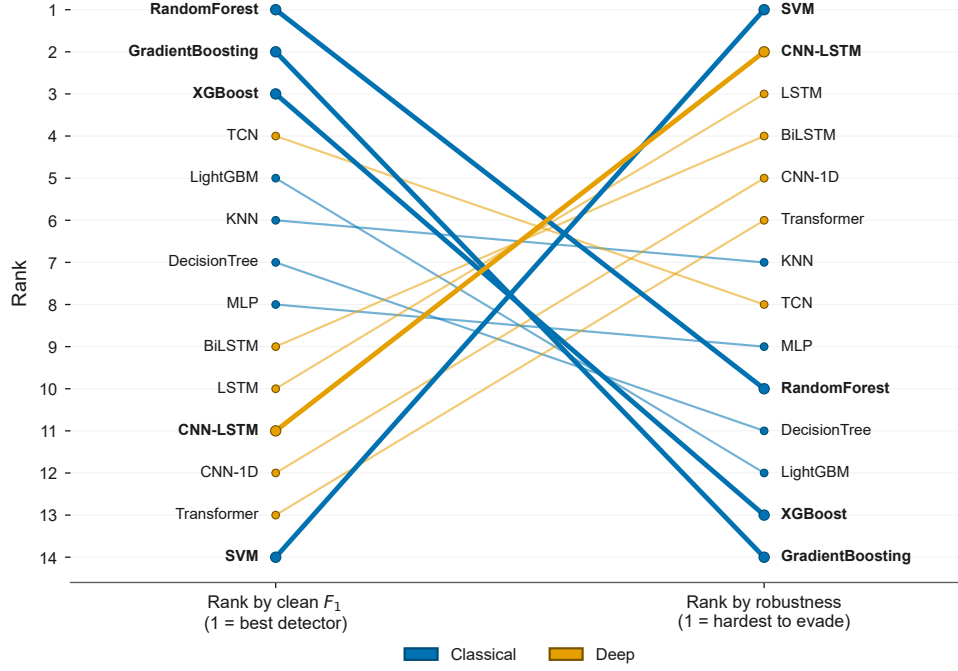


Fig. 14 Detectors ranked by clean F_1 against the same detectors ranked by resistance to the decision based attack. The two orderings run close to opposite, which is the same relation the transmitted attacks produce.

reference to any detector, it cannot be attributed to adversarial optimisation against a particular model.

Where resistance does appear, it is purchased with false alarms. Five detectors flag every one of the 126 attacks, and those same five flag between 46 and 74 percent of authentic samples. A receiver alarming on half of its authentic observations would be disabled in service, so this resistance is not operationally available, and the pattern is the accuracy against robustness trade of Section 2.4 appearing in a GNSS setting. Figure 15 shows the same point across the full set of criteria a deployment decision weighs: no detector dominates on all of them, and the orderings induced by different criteria disagree, so no single scalar ranks the set.

A missed detection is consequential rather than benign. The attack propagated through the navigation stage drove the receiver clock at the rate the transmitter was configured to impose, to within 0.05 percent, while the detector with the best clean performance classified 98.8 percent of its samples as authentic. Detection failure and navigation domain effect were measured on the same attack and the same recording, so the two are not independent observations juxtaposed after the fact.

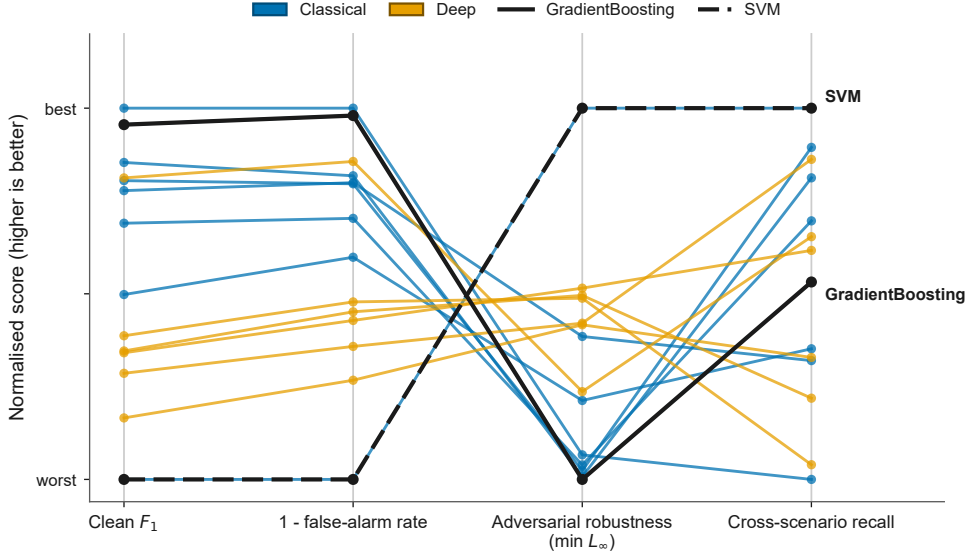


Fig. 15 Every detector across the criteria a deployment decision would weigh, each normalised so higher is better. No detector is best on all of them, and the orderings disagree, so a single accuracy figure cannot rank them.

7.2 Implications for detector evaluation

Evaluating at the waveform level fixes the interpretation of the resulting failure rates. A displacement applied directly to a table of receiver observables corresponds to an attack only if some incident waveform produces that displacement. Several observables a receiver reports are computed from the same correlator accumulations and cannot be varied independently, so an evaluation that varies them independently may report vulnerability to a receiver state that no transmission realises. When the attack is generated as a signal, the observables are by construction whatever the receiver computed while tracking it, and every constraint receiver processing imposes is satisfied because that processing imposed it.

Section 6.5 quantifies what this changes. For the tree based detectors the feature space and waveform level evaluations agree, and the fragility those models exhibit is a property of the models rather than of either evaluation. For the support vector machine the two disagree entirely: its minimum perturbation is the largest in the study, which taken alone would identify it as the most robust detector, yet it is evaded by 53 of the 126 transmitted attacks. Its false alarm rate of 0.960 reconciles the two readings, since a classifier that flags almost everything is expensive to cross in feature space while remaining useless as a monitor. The conclusion is not that one evaluation is invalid, but that a perturbation magnitude reported without the false alarm rate that accompanies it can invert the ordering it appears to establish.

A related limitation of accuracy based evaluation appears under distribution shift. Figure 16 reports recall on data withheld from training, holding out either an entire attack scenario or individual satellites. Recall falls in both cases, which indicates

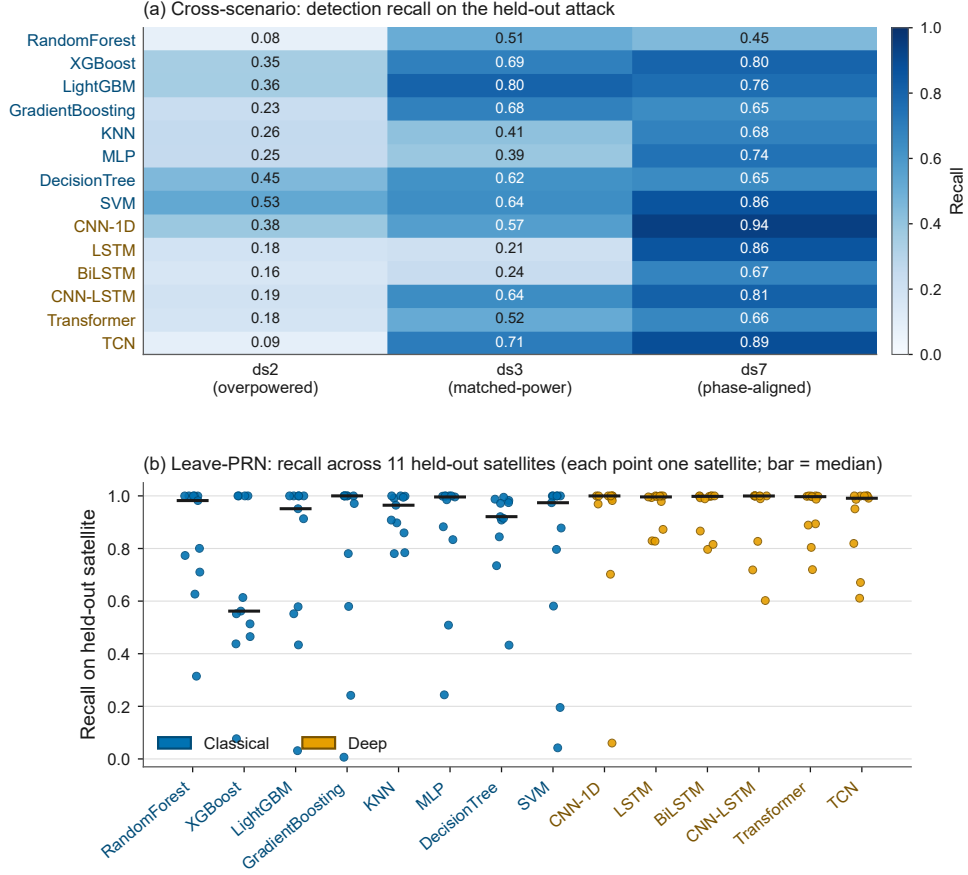


Fig. 16 Detection recall on data held out from training. Panel (a) holds out a whole attack scenario and panel (b) holds out satellites one at a time. Recall falls in both, so the detectors depend on conditions present in their training data.

that the detectors depend on conditions present in their training data even before an adversary is introduced.

7.3 Implications for receiver integrity

The transmitter settings evaluated here were selected without reference to any detector, and 126 of them capture a receiver. Approximately two fifths of those evade the strongest detector tested, and none of that required knowledge of the detector, access to its gradients, or the ability to query it. An adversary possessing any of those capabilities would be strictly stronger.

For a receiver integrity monitor the practical consequence is that detection quality measured on a fixed corpus is not a security property. It characterises behaviour on that corpus, and a detector intended to resist an adversary must be characterised

against signals the adversary can select and emit, with the false alarm rate reported alongside.

The false alarm rates observed here carry a further implication independent of any adversary. At a recall floor of 0.95 they span 0.337 to 0.960, so no detector in this study is usable as a standalone single epoch integrity monitor even in the absence of an attacker. A deployed monitor would necessarily combine decisions across epochs and across satellites, trading detection latency against alarm rate, and the per epoch quantities reported here are the inputs to such a scheme rather than a substitute for it.

7.4 Scope and limitations

The adversary is restricted in the four respects set out in Section 4, and each restriction reduces its capability, so the failure rates reported are a lower bound on what an unrestricted adversary could achieve.

The study covers one signal, GPS L1 C/A, and one stationary receiver at a surveyed location. Whether the same detectors behave equivalently on other civil signals, or on a moving platform whose dynamics enter the observables, is not established here. The generator itself is not specific to GPS L1 C/A, and repeating the procedure on another signal requires the corresponding spreading code and receiver configuration.

The navigation domain result rests on a common pull-off applied to all satellites, which a stationary receiver absorbs into its clock estimate. An adversary seeking to displace the reported position rather than the clock would apply differing offsets across satellites, and that case is not measured here.

One documented difference between the generated attack and the recorded one is reported in Section 5.3. The generated counterfeit is an amplitude matched replica, so its cancellation is cleaner than the recorded spoofer achieved, leaving the generated attack marginally stronger and therefore marginally easier to detect. This bias acts against the adversary and so does not inflate the failure rates reported.

7.5 Directions for a defence

The results constrain what a defence can rely upon. Architecture is not sufficient, since both model families appear among the detectors that are evaded. Clean detection quality is not sufficient, since the detector with the best clean performance is evaded most often. Threshold placement is not sufficient, since resistance obtained by raising the alarm rate is paid for on authentic data.

Two directions remain open. The first is to require the detector to respect a physical relation that authentic signals satisfy, as has been done for code carrier consistency (Song et al. 2026); the library of transmitted attacks constructed here supplies the signals against which such a detector would need to be evaluated, including the settings that hold received power constant through Equation 2. The second is to construct a detector whose behaviour under bounded input change can be certified in advance rather than estimated after the fact (Cohen et al. 2019). Both lie outside the scope of the present study, which characterises the behaviour of existing detectors.

8 Conclusion

This paper has characterised the behaviour of machine learning detectors of GNSS spoofing under attacks that are transmitted signals selected by an adversary. Each attack was synthesised as radio samples, summed with authentic recordings from the Texas Spoofing Test Battery, passed through a model of the receiver front end and tracked by an open software receiver, so that the observables each detector read were produced by receiver processing of a transmittable waveform. The generator was validated against a recorded attack whose construction is documented in full, and reproduced its observables satellite by satellite.

Fourteen detectors, eight classical and six deep, were placed at a common operating point defined by a recall floor of 0.95 on clean validation data, and evaluated against 126 transmitter settings satisfying the published conditions for receiver takeover.

Clean detection performance was found to be a poor predictor of resistance, and at the top of the ranking the two orderings were close to inverted. The random forest attained the highest clean F_1 score at 0.833 and the lowest false alarm rate at 0.337, and failed to flag 55 of the 126 attacks, flagging 0.1 percent of the spoofed samples of one of them. The five detectors that flagged every attack did so while flagging between 46 and 74 percent of authentic samples, a regime in which the resistance is not operationally available. Across all fourteen detectors, false alarm rates at this operating point ranged from 0.337 to 0.960.

One evaded attack was propagated through the navigation stage. It drove the receiver clock at 48.89 nanoseconds per second against a commanded 48.87, while the detector with the best clean performance classified 98.8 percent of its samples as authentic, establishing detection failure and navigation domain effect on the same attack and the same recording.

The adversary producing these results had no knowledge of the deployed detector, computed no gradient and issued no query, and was further restricted to a single signal, full satellite coverage and a stationary victim. Each restriction reduces its capability, so the reported failure rates constitute a lower bound.

The broader implication is that detection accuracy measured on a fixed corpus of recorded attacks characterises that corpus. A detector intended to withstand an adversary requires characterisation against signals the adversary can select and emit, reported jointly with the false alarm rate incurred. The library of transmitted attacks constructed here is available for that purpose, and extends directly to the evaluation of physics constrained and certified detectors.

List of abbreviations

CNN convolutional neural network
C/N₀ carrier-to-noise density ratio
DLL delay-lock loop
FGI Finnish Geospatial Research Institute
FLL frequency-lock loop
GNSS Global Navigation Satellite System
GPS Global Positioning System

LSTM long short-term memory
PLL phase-lock loop
PNT positioning, navigation and timing
PRN pseudo-random noise
TEXBAT Texas Spoofing Test Battery

Declarations

Ethics approval and consent to participate. Not applicable. This study involves no human participants, human data or animal subjects; it processes recorded radio-frequency signals from a public spoofing-attack test benchmark.

Consent for publication. Not applicable.

Availability of data and materials. The derived receiver-observable corpus and the evaluation code are available from the corresponding author on reasonable request. The raw Texas Spoofing Test Battery recordings are distributed by the University of Texas at Austin Radionavigation Laboratory, and the FGI-GSRx software-defined receiver is released under the GPLv3 licence.

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References

- Akos DM (2012) Who's afraid of the spoofer? GPS/GNSS spoofing detection via automatic gain control (AGC). *Navigation* 59(4):281–290. <https://doi.org/10.1002/navi.19>
- An S, Jang DJ, Lee EK (2025) Adversarial evasion attacks on SVM-based GPS spoofing detection systems. *Sensors* 25(19):Art. no. 6062. <https://doi.org/10.3390/s25196062>
- Athalye A, Carlini N, Wagner D (2018) Obfuscated gradients give a false sense of security. In: *Proc. ICML, Stockholm, Sweden*, pp 274–283, <https://doi.org/10.48550/arXiv.1802.00420>
- Bai S, Kolter JZ, Koltun V (2018) An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv:1803.01271* <https://doi.org/10.48550/arXiv.1803.01271>
- Bhatti J, Humphreys TE (2017) Hostile control of ships via false GPS signals: Demonstration and detection. *NAVIGATION* 64(1):51–66. <https://doi.org/10.1002/navi.183>

- Biggio B, Roli F (2018) Wild patterns: Ten years after the rise of adversarial machine learning. *Pattern Recognition* 84:317–331. <https://doi.org/10.1016/j.patcog.2018.07.023>
- Borhani-Darian P, Li H, Wu P, et al (2024) Detecting GNSS spoofing using deep learning. *EURASIP J Adv Signal Process* 2024:Art. no. 14. <https://doi.org/10.1186/s13634-023-01103-1>
- Breiman L (2001) Random forests. *Mach Learn* 45(1):5–32. <https://doi.org/10.1023/A:1010933404324>
- Brendel W, Rauber J, Bethge M (2018) Decision-based adversarial attacks: Reliable attacks against black-box machine learning models. In: *Proc. ICLR*, <https://doi.org/10.48550/arXiv.1712.04248>
- Cao Y, et al (2021) Invisible for both camera and lidar: Security of multi-sensor fusion based perception in autonomous driving under physical-world attacks. In: *Proc. IEEE S&P*, San Francisco, CA, USA, pp 176–194, <https://doi.org/10.1109/SP40001.2021.00076>
- Carlini N, Wagner D (2018) Audio adversarial examples: Targeted attacks on speech-to-text. In: *Proc. IEEE S&P Workshops*, San Francisco, CA, USA, pp 1–7, <https://doi.org/10.1109/SPW.2018.00009>
- Carlini N, Athalye A, Papernot N, et al (2019) On evaluating adversarial robustness. *arXiv preprint arXiv:1902.06705*
- Chen J, Jordan MI, Wainwright MJ (2020) HopSkipJumpAttack: A query-efficient decision-based attack. In: *Proc. IEEE Symp. Security and Privacy (S&P)*, pp 1277–1294, <https://doi.org/10.1109/SP40000.2020.00045>
- Chen L, Ouyang X, Zeng F, et al (2025) GNSS spoofing detection based on lightweight features and CGAN-ANN in unknown scenarios. *GPS Solutions* 29(4):Art. no. 175. <https://doi.org/10.1007/s10291-025-01938-1>
- Chen T, Guestrin C (2016) XGBoost: A scalable tree boosting system. In: *Proc. ACM KDD*, San Francisco, CA, USA, pp 785–794, <https://doi.org/10.1145/2939672.2939785>
- Chen Z, Li J, Li J, et al (2022) GNSS multiparameter spoofing detection method based on support vector machine. *IEEE Sens J* 22(18):17864–17874. <https://doi.org/10.1109/JSEN.2022.3193388>
- Cohen JM, Rosenfeld E, Kolter Z (2019) Certified adversarial robustness via randomised smoothing. In: *Proc. ICML*, Long Beach, CA, USA, pp 1310–1320, <https://doi.org/10.48550/arXiv.1902.02918>

- Finnish Geospatial Research Institute (2022) FGI-GSRx open-source multi-GNSS software-defined receiver. National Land Survey of Finland, gPLv3. <https://github.com/nlsfi/FGI-GSRx>
- Friedman JH (2001) Greedy function approximation: A gradient boosting machine. *Ann Stat* 29(5):1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Goodfellow IJ, Shlens J, Szegedy C (2015) Explaining and harnessing adversarial examples. In: Proc. ICLR, San Diego, CA, USA, <https://doi.org/10.48550/arXiv.1412.6572>
- Hochreiter S, Schmidhuber J (1997) Long short-term memory. *Neural Comput* 9(8):1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Humphreys TE (2016) TEXBAT data sets 7 and 8. Tech. rep., Radionavigation Laboratory, University of Texas at Austin, available: <https://rnl-data.ae.utexas.edu/datastore/texbat/>
- Humphreys TE, Bhatti JA, Shepard DP, et al (2012) The Texas spoofing test battery: Toward a standard for evaluating GPS signal authentication techniques. In: Proc. ION GNSS, Nashville, TN, USA, pp 3569–3583
- Iqbal A, Aman MN, Sikdar B (2024) A deep learning based induced GNSS spoof detection framework. *IEEE Trans Mach Learn Commun Netw* 2:457–478. <https://doi.org/10.1109/TMLCN.2024.3386649>
- Jullian O, Otero B, Stojilović M, et al (2022) Deep learning detection of GPS spoofing. In: Machine Learning, Optimization, and Data Science (LOD 2021), Lecture Notes in Computer Science, vol 13163. Springer, Cham, Switzerland, https://doi.org/10.1007/978-3-030-95467-3_38
- Kaplan ED, Hegarty CJ (2005) Understanding GPS: Principles and Applications, 2nd edn. Artech House, Norwood, MA
- Ke G, et al (2017) LightGBM: A highly efficient gradient boosting decision tree. In: Proc. NeurIPS, Long Beach, CA, USA, pp 3146–3154
- Kerns AJ, Shepard DP, Bhatti JA, et al (2014) Unmanned aircraft capture and control via GPS spoofing. *J Field Robot* 31(4):617–636. <https://doi.org/10.1002/rob.21513>
- Li J, Chen Z, Yuan X, et al (2025) A real-time GNSS time spoofing detection framework based on feature processing. *GPS Solutions* 29(1):45. <https://doi.org/10.1007/s10291-024-01802-8>
- Lin G, et al (2023) Boosting adversarial transferability with shallow-feature attack on SAR images. *Remote Sens* 15(10):Art. no. 2699. <https://doi.org/10.3390/rs15102699>

- Madry A, Makelov A, Schmidt L, et al (2018) Towards deep learning models resistant to adversarial attacks. In: Proc. ICLR, Vancouver, BC, Canada, <https://doi.org/10.48550/arXiv.1706.06083>
- Mahroof A, Farooq SZ, Nabi I, et al (2024) Machine learning-based detection of spoofing attacks in GNSS: A study using TEXBAT dataset. In: Proc. 14th Int. Conf. Electrical Engineering (ICEENG), Cairo, Egypt, pp 1–6, <https://doi.org/10.1109/ICEENG58856.2024.10566287>
- Petit J, et al (2015) Remote attacks on automated vehicles sensors: Experiments on camera and lidar. In: Proc. Black Hat Europe, Amsterdam, Netherlands, pp 1–11
- Psiaki ML, Humphreys TE (2016) GNSS spoofing and detection. Proc IEEE 104(6):1258–1270. <https://doi.org/10.1109/JPROC.2016.2526658>
- Romaniuc AG, Vasile VC, Borda ME (2025) Global Navigation Satellite System spoofing attack detection using receiver independent exchange format data and Long Short-Term Memory algorithm. Information 16(6):Art. no. 502. <https://doi.org/10.3390/info16060502>
- Song Y, Xiao W, Ye X, et al (2026) Generalized GNSS spoofing detection in unseen scenarios using physics-constrained neural networks with code-carrier consistency. Satellite Navigation 7(1):15. <https://doi.org/10.1186/s43020-026-00199-8>
- Sung YH, Park SJ, Kim DY, et al (2022) GPS spoofing detection method for small UAVs using 1D convolution neural network. Sensors 22(23):Art. no. 9412. <https://doi.org/10.3390/s22239412>
- Szegedy C, et al (2014) Intriguing properties of neural networks. In: Proc. ICLR, Banff, AB, Canada, <https://doi.org/10.48550/arXiv.1312.6199>
- Tsipras D, Santurkar S, Engstrom L, et al (2019) Robustness may be at odds with accuracy. In: Proc. ICLR, New Orleans, LA, USA, <https://doi.org/10.48550/arXiv.1805.12152>
- Vaswani A, et al (2017) Attention is all you need. In: Proc. NeurIPS, Long Beach, CA, USA, pp 5998–6008, <https://doi.org/10.48550/arXiv.1706.03762>
- Yang M, Oracevic A, Dilek S (2025) Deep learning-based defense against GPS spoofing attacks in unmanned aerial vehicles. In: Proc. Int. Conf. Smart Applications, Communications and Networking (SmartNets), Istanbul, Türkiye, <https://doi.org/10.1109/SmartNets65254.2025.11106869>